

Section 4: Non-Sobolev Compactness

1 Non-Sobolev Compactness

1.1 Compactness Framework

To overcome limitations of classical Sobolev embeddings, we define the solution space H using a scale-invariant norm:

$$\|u\|_H = \|u\|_{L^2} + \|\nabla u\|_{L^2} + \sup_{k>k_0} \frac{E(k)}{k^\beta}, \quad \beta > 0,$$

where:

- $\|u\|_{L^2}$ is the total kinetic energy.
- $\|\nabla u\|_{L^2}$ measures enstrophy.
- $\sup_{k>k_0} \frac{E(k)}{k^\beta}$ controls high-frequency energy contributions, with β ensuring rapid spectral decay.

This definition captures scale-invariant properties of the flow and bypasses reliance on classical Sobolev embeddings, which are often insufficient for controlling high-frequency behaviors in turbulent flows. The norm H is particularly suited for flows where classical compactness arguments fail due to spectral energy cascades.

Comparison with Classical Sobolev Spaces: Unlike traditional Sobolev embeddings (e.g., H^1 or H^2), the H -norm explicitly incorporates high-frequency energy suppression via $\sup_{k>k_0} \frac{E(k)}{k^\beta}$. This ensures that the compactness framework remains robust even in the presence of turbulent cascades, where classical embeddings may break down.

Physical Interpretation: The H -norm reflects key physical properties of turbulent flows:

- ****Low-Frequency Components:**** The terms $\|u\|_{L^2}$ and $\|\nabla u\|_{L^2}$ capture bulk energy and enstrophy, representing large-scale flow structures.
- ****High-Frequency Suppression:**** The term $\sup_{k>k_0} \frac{E(k)}{k^\beta}$ quantifies the damping of small-scale energy transfer, preventing blowup.

1.2 Compactness Proof

We prove strong convergence in L^2 for approximate solutions in H without relying on classical Sobolev embeddings.

Theorem 1.1 (Compactness in H). *Let $\{u_n\} \subset H$ be a sequence of approximate solutions satisfying:*

1. $\sup_n \|u_n\|_H < \infty$.
2. $\partial_t u_n$ is uniformly bounded in $L^2(0, T; H^{-1})$.

Then, $\{u_n\}$ has a subsequence that converges strongly in $L^2(0, T; L^2)$.

Proof. The proof adapts the Aubin–Lions lemma while leveraging spectral decay bounds instead of Sobolev embeddings. We proceed in four steps.

Step 1: Uniform Boundedness in H . The assumption $\sup_n \|u_n\|_H < \infty$ implies:

- $\{u_n\}$ is uniformly bounded in $L^2(0, T; L^2)$.
- $\{\nabla u_n\}$ is uniformly bounded in $L^2(0, T; L^2)$.
- High-frequency contributions $\sup_{k > k_0} \frac{E_n(k)}{k^\beta}$ are uniformly controlled.

This boundedness ensures that u_n belongs to the solution space H for all n .

Step 2: Weak Compactness in Time. The boundedness of $\partial_t u_n$ in $L^2(0, T; H^{-1})$ implies that u_n is weakly compact in $L^2(0, T; L^2)$. Specifically, there exists a subsequence (still denoted u_n) and a limit function $u \in L^2(0, T; L^2)$ such that:

$$u_n \rightharpoonup u \quad \text{in } L^2(0, T; L^2).$$

Step 3: Spectral Decay and Approximation. From the Scale-Barrier Principle, the spectral energy satisfies:

$$E(k) \leq Ck^{-\gamma}, \quad \gamma > \beta + 2.$$

This implies that for any $\epsilon > 0$, there exists a sufficiently large k_0 such that:

$$\int_{k > k_0} E(k) dk < \epsilon.$$

High-frequency components are thus negligible beyond a critical wave number k_0 . Using this spectral decay, we approximate u_n by smooth, compactly supported functions in Fourier space, ensuring that:

$$\|u_n - u_n^{(k_0)}\|_{L^2} < \epsilon,$$

where $u_n^{(k_0)}$ represents the low-pass filtered version of u_n with frequencies restricted to $k \leq k_0$.

Step 4: Strong Convergence in L^2 . Combining weak compactness from Step 2 with the approximation result from Step 3, we conclude that:

$$\|u_n - u\|_{L^2(0,T;L^2)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

The strong convergence follows from the control of high-frequency components and weak compactness in the low-frequency regime. $\square$

1.3 Significance

Compactness in the space H ensures smooth limits for approximate solutions and plays a critical role in establishing the existence and regularity of global solutions to the Navier–Stokes equations. Key implications include:

1. ****Spectral Control:**** The scale-invariant norm suppresses high-frequency energy cascades, ensuring convergence without relying on classical Sobolev embeddings.
2. ****Global Regularity:**** Strong convergence in L^2 supports the passage to smooth limits in weak formulations.
3. ****Applicability:**** The framework extends naturally to fractional Sobolev norms and bounded domains, allowing broader applicability to real-world flows.

Applications in Numerical Analysis: The compactness framework is particularly advantageous for numerical simulations. By ensuring strong convergence, the framework:

- Guarantees stability in iterative schemes for solving Navier–Stokes equations.
- Provides error bounds for approximate solutions based on spectral decay properties.
- Enhances the reliability of turbulence models in capturing high-frequency dynamics.

References

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